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Ideal stability of cylindrical plasma in the presence of mass flow

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The ideal stability of cylindrical plasma with mass flows is investigated using the guiding center plasma (GCP) model of Grad [*Proceedings of the Symposium on Electromagnetic and Fluid Dynamics of Gaseous Plasmas* (Polytechnic Inst. of Brooklyn, New York, 1961), p. 37]. For rotating plasmas, the kinetic treatment of the parallel motion in GCP gives significantly different results from the fluid models, where the pressures are obtained from equations of state. In particular, GCP removes the resonance with slow magnetoacoustic waves and the loss of stability that occurs in magnetohydrodynamics (MHD) for near-sonic flows. Because of the strong kinetic damping of the sound waves in an isothermal plasma, the slow waves have little influence on plasma stability in GCP at low beta. In the large aspect ratio, low-beta tokamak ordering, Alfvénic flows are needed to change the ideal GCP stability significantly. At lowest order in the inverse aspect ratio, flow can be favorable or unfavorable for stability of local modes depending on the profiles, but external kinks are always destabilized by flow if the velocity vanishes at the edge. For high-beta, reversed field pinch equilibria, numerical computations show that flow can be stabilizing for local modes, but external modes are destabilized by flow. In three dimensions, the MHD equilibrium problem becomes hyperbolic for arbitrarily small flows across the magnetic field, whereas the GCP equilibrium equation remains elliptic for sub-Alfvénic flows.

I. INTRODUCTION

In contrast with static plasma equilibria, where a detailed picture has been established for the macroscopic stability of various devices, little is known at present about the stability of equilibria with mass flows. Nevertheless, the stability of rotating plasmas has become a question of practical importance in fusion research because of the substantial neutral beam power in recent tokamak experiments,¹⁻³ available not only to heat the plasma, but also to produce mass flows. Furthermore, plasma flows are ubiquitous and frequently play important roles in astrophysical and geophysical contexts.^{4,5}

In this paper, we investigate the ideal linear stability of cylindrical equilibria with arbitrary profiles of magnetic fields and mass flow. Previous studies⁶⁻⁸ based on magnetohydrodynamics (MHD) predicted that flows of the order of the sound speed would lead to complete loss of stability because of resonance with slow magnetoacoustic waves. However, there is experimental evidence that the sound speed does not present a stability limit, e.g., from the F-II device in Stockholm.⁹ In addition, it is well known¹⁰ that the MHD equations poorly describe the slow magnetoacoustic and ion-sound waves by assuming an equation of state, $p \propto \rho^\gamma$. A semimacroscopic theory better suited for the description of rotating plasmas is the guiding center plasma (GCP) model of Grad.¹¹ The GCP model is identical to MHD for plasma motion across the magnetic field, but uses kinetic theory to describe the motion of particles along the field lines. It introduces Landau damping of the sound waves, which strongly reduces their effect on plasma stability, and indeed removes the MHD stability limit at the sound speed.

In Sec. II, we derive the second-order radial eigenvalue problem, including the effects of mass flow for several differ-

ent models: GCP, MHD (including a parallel viscosity), and the double adiabatic theory of Chew, Goldberger, and Low¹² (CGL). The difference between these descriptions lies in three coefficients, expressing the response in plasma density and perpendicular pressure to perturbations of the parallel magnetic field and to radial displacements. In the inviscid fluid models, the plasma response shows a resonance when the parallel phase speed of the mode in the local center of mass frame equals the sound speed. This resonance is removed in GCP by the kinetic treatment of the parallel motion, and the influence of the magnetoacoustic wave on the stability of rotating plasmas is much reduced. As a consequence, the sound speed does *not* set a stability limit in GCP. (The resonance is also removed by adding parallel viscosity to the MHD equations, as was done in a recent calculation by Gimblett.¹³ However, for frequencies larger than the ion-ion collision frequency, the response of the viscous MHD plasma is quite different from that of GCP.) An interesting result is that at *marginal stability* and in the *absence* of flow, the response of the guiding center model for a plasma in local thermodynamic equilibrium is identical to that of inviscid MHD with the adiabatic index γ set to unity, and differs from the double-adiabatic approximation. As a consequence, MHD and GCP predict the same marginal points for *static* equilibria, whereas CGL is more optimistic.

In Sec. III, we consider the stability of local modes and find that whenever the flow has shear, the GCP description leads to a complex Suydam index. This implies that the infinite sequences of unstable modes occurring in MHD when the Suydam criterion is violated do not generally exist in GCP (except for static equilibria, as shown by Pao¹⁴). For an equilibrium of reversed field pinch (RFP) type, we find numerically that flow is stabilizing for the localized modes, due, in part, to the kinetic damping in GCP. However, in the

limit of low-beta, large aspect ratio tokamaks, the flow must be Alfvénic to give an $O(1)$ contribution to the Suydam index. This contribution can be stabilizing only if the flow has a pitch opposite to that of the magnetic field.

In Sec. IV, we consider global modes, in particular external kinks, and use the "straight tokamak" or lowest-order reduced-MHD ordering to study the effects of Alfvénic flows. To lowest order in inverse aspect ratio ϵ , the slow waves are completely eliminated in GCP. An integral expression, similar to δW , is derived for modes locked to resistive walls. This expression shows that any flow that vanishes at the plasma surface is destabilizing for wall-locked external kink modes. Our numerical computations confirm this conclusion also for RFP equilibria and for the cylindrical internal kink.

In Sec. V, we discuss properties of the equilibrium problem in the presence of flow by applying the linearized equations to perturbations with zero frequency in the laboratory frame. As is well known,¹⁵ the MHD Grad-Shafranov equation can become hyperbolic because of mass flow, and there exists a "first hyperbolic region" where the poloidal flow is close to the poloidal sound speed. We show that, in three dimensions, the introduction of *arbitrarily* small flows perpendicular to the magnetic field makes the MHD equilibrium equation hyperbolic. However, the guiding center plasma has very different properties, and the equilibrium equation remains elliptic for sub-Alfvénic flows. Dobrott and Greene¹⁶ noted the pronounced difference between the fluid and kinetic models in the case of axisymmetric equilibrium. The absence of the first hyperbolic region in the GCP equilibrium problem, as well as the absence of pronounced degradation of stability for sonic flows, are both results of the near elimination of the slow magnetoacoustic wave in a kinetic description.

II. THE RADIAL EIGENVALUE PROBLEM

It is well known¹⁰ that MHD does not properly describe slow magnetoacoustic waves in collisionless, isothermal plasma, and that the equation of state, $p \propto \rho^\gamma$, can only be justified in the limit of large collisionality, $\nu_{ii} \gg |\tilde{\omega}|$, where $\tilde{\omega}$ stands for the wave frequency in the local center of mass frame. The double adiabatic theory of Chew, Goldberger, and Low¹² lacks another important piece of physics by ignoring the heat flow along the field lines, and only holds in the limit of large phase velocity, $|\tilde{\omega}| \gg \max(k_{\parallel} v_{the}, \nu_{ii})$. Similar to MHD, it fails to produce damping of the ion-sound waves. As shown in Ref. 7, ion-sound waves play a prominent role for the MHD stability of rotating plasmas. Consequently, a more accurate description of the parallel dynamics is called for, and we find that the GCP model of Grad¹¹ is a more adequate tool for the flow problem. GCP uses kinetic theory for the motion along the field lines and assumes no ordering of the parallel phase velocity with respect to the thermal velocities. The slow magnetosonic wave has very different properties in MHD and GCP,¹⁰ and in the kinetic description, the magnetosonic wave is almost eliminated in isothermal plasma. In the present paper, we show that this modification of the slow wave physics has significant consequences for the stability of plasmas with sheared flows, viz.,

the MHD instability induced by resonance with slow waves⁷ is removed and with it, the unconditional instability for near-sonic flows.

We now derive the second-order cylindrical eigenvalue problem¹⁷ for GCP, previously stated in Ref. 18. We first treat the fluid part of the calculation, leaving the pressures unspecified. This allows us to substitute either the MHD, CGL, or GCP model for the parallel motion and compare the results simply by substituting different forms for three coefficients into the second-order radial equation. The same approach is taken in Ref. 19, where we discuss the problem of toroidal equilibria with flow. The reader not interested in details may skip the derivation in Secs. II A and B.

A. Fluid calculation

The linearized MHD equations for equilibria with mass flow were given by Frieman and Rotenberg.²⁰ We generalize their model by allowing for anisotropic pressure:

$$\rho \frac{\partial^2 \xi}{\partial t^2} + 2\rho \mathbf{v} \cdot \nabla \frac{\partial \xi}{\partial t} = -\nabla \cdot (\mathbf{P} + |\mathbf{B} \cdot \mathbf{Q}|) + \mathbf{B} \cdot \nabla \mathbf{Q} + \mathbf{Q} \cdot \nabla \mathbf{B} + \nabla \cdot [\rho \xi (\mathbf{v} \cdot \nabla) \mathbf{v} - \rho \mathbf{v} (\mathbf{v} \cdot \nabla) \xi], \quad (1)$$

$$\mathbf{P} = \{p_{\perp} | + [(p_{\parallel} - p_{\perp})/B^2] \mathbf{B} \mathbf{B}\}_1. \quad (2)$$

Here, ξ is the Lagrangian displacement, the subscript "1" denotes Eulerian perturbed quantities, and $\mathbf{Q} = \mathbf{B}_1$. In the following, we make extensive use of Lagrangian perturbations, such as

$$\delta \mathbf{v} = \mathbf{v}_1 + \xi \cdot \nabla \mathbf{v} = \left(\frac{\partial}{\partial t} + \mathbf{v} \cdot \nabla \right) \xi, \quad (3a)$$

$$\delta \rho = \rho_1 + \xi \cdot \nabla \rho = -\rho \nabla \cdot \xi, \quad (3b)$$

$$\delta \mathbf{B} = \mathbf{Q} + \xi \cdot \nabla \mathbf{B} = \mathbf{B} \cdot \nabla \xi - \mathbf{B} \nabla \cdot \xi. \quad (3c)$$

The equilibrium relation in a cylinder reads

$$\frac{d}{dr} \left(p_{\perp} + \frac{B^2}{2} \right) = \frac{(\rho v_{\theta}^2 - \sigma B_{\theta}^2)}{r}, \quad (4a)$$

where

$$\sigma = 1 - \Delta = 1 - (p_{\parallel} - p_{\perp})/B^2 \quad (4b)$$

is the anisotropy factor. Assuming an $\exp[i(\omega t + m\theta - kz)]$ dependence for the perturbed variables, we obtain from (1) using (3b) and (3c)

$$\begin{aligned} A \xi + \hat{r} \left[r \xi_r \frac{d}{dr} \left(\frac{\sigma B_{\theta}^2 - \rho v_{\theta}^2}{r^2} \right) - \frac{\delta \rho}{\rho} \frac{\sigma B_{\theta}^2 - \rho v_{\theta}^2}{r} \right] \\ + \frac{2iT}{r} \hat{z} \times \xi \\ = \nabla p_{\star} - \left(iF\mathbf{B} - \frac{\hat{r} B_{\theta}^2}{r} \right) \left(\delta \sigma + \frac{\sigma \delta \rho}{\rho} \right), \end{aligned} \quad (5)$$

where

$$p_{\star} = p_{1\perp} + \mathbf{B} \cdot \mathbf{Q} \quad (6)$$

is the perturbation in total perpendicular pressure and $\hat{r}$ and $\hat{z}$ denote unit vectors in the r and z directions, respectively. The coefficients in Eq. (5) are given by

$$A = \rho \tilde{\omega}^2 - \sigma F^2, \quad (7a)$$

$$T = \sigma F B_{\theta} - \rho \tilde{\omega} v_{\theta}, \quad (7b)$$

with

$$F = \mathbf{k} \cdot \mathbf{B} = k_{\parallel} B = m B_{\theta} / r - k B_z, \quad (7c)$$

and where

$$\tilde{\omega} = \omega + \mathbf{k} \cdot \mathbf{v} = \omega + m v_{\theta} / r - k v_z \quad (7d)$$

is the Doppler shifted frequency. In Eq. (5), all information about the parallel dynamics is contained in $\delta\rho, p_*$, and $\delta\sigma$. The perturbed magnetic field has been eliminated except for the parallel component δB appearing in $\delta\sigma$.

The displacement is decomposed according to

$$\xi = \xi_r \mathbf{r} + \xi_{\perp} \frac{\hat{\mathbf{r}} \times \mathbf{B}}{B} + \xi_{\parallel} \frac{\mathbf{B}}{B}. \quad (8)$$

Simple consequences of (3b) and (3c) are

$$B \delta B = i E B \xi_{\perp} - \frac{B^2}{r} \frac{d}{dr} (r \xi_r) + \frac{B_{\theta}^2}{r} \xi_r, \quad (9a)$$

$$\frac{\delta B}{B} - \frac{\delta\rho}{\rho} = i k_{\parallel} \xi_{\parallel} + \frac{B_{\theta}^2}{r B^2} \xi_r, \quad (9b)$$

with

$$E = m B_z / r + k B_{\theta} = \hat{\mathbf{r}} \cdot (\mathbf{k} \times \mathbf{B}) = k_{\perp} B. \quad (10)$$

The equation of motion, written in component form, reads

$$\begin{aligned} & \left[A + r \frac{d}{dr} \left(\frac{\sigma B_{\theta}^2 - \rho v_{\theta}^2}{r^2} \right) + \frac{B_{\theta}^2}{r^2 B^2} \frac{2 T B_{\theta}}{F} \right] \xi_r \\ & + \frac{\delta\rho}{r} \left(X^2 - \frac{\tilde{\omega}^2 B_{\theta}^2}{F^2} \right) + \frac{2 i T B_z}{r B} \xi_{\perp} \\ & = \frac{d p_*}{dr} + \frac{2 T B_{\theta}}{r F} \frac{\delta B}{B} \\ & + \frac{B_{\theta}^2}{r B^2} [\delta p_{\perp} - \delta p_{\parallel} + 2 \Delta B \delta B], \end{aligned} \quad (11)$$

$$A B \xi_{\perp} = (2 i T B_z / r) \xi_r - i E p_*, \quad (12)$$

$$\begin{aligned} & \rho \tilde{\omega}^2 (i F B \xi_{\parallel} + (B_{\theta}^2 / r) \xi_r) \\ & = F^2 [\Delta B \delta B - \delta p_{\parallel} + (\rho X^2 / r) \xi_r], \end{aligned} \quad (13)$$

where

$$X = v_{\theta} - \tilde{\omega} B_{\theta} / F \quad (14)$$

is related to the crossfield flow of the equilibrium. Substituting (9b) in the parallel momentum equation (13), we can eliminate $\xi_{\parallel}$ and obtain a relation between the perturbations in parallel pressure, density, and magnetic field:

$$k_{\parallel}^2 \left(\delta p_{\parallel} - \Delta B \delta B - \frac{\rho X^2}{r} \xi_r \right) = \rho \tilde{\omega}^2 \left(\frac{\delta\rho}{\rho} - \frac{\delta B}{B} \right). \quad (15)$$

B. Parallel dynamics

To close the system (11)–(13), we must express $\delta\rho$ and $\delta p_{\perp}$ in terms of δB and ξ_r . In the MHD and double adiabatic equations, this is accomplished by substituting the respective equations of state in the parallel equation of motion (15), as shown in Appendix B. In Grad's guiding center model,¹¹ the motion along the field lines is treated kinetically. All particles are assumed to have the same velocity $\mathbf{U} = \mathbf{E} \times \mathbf{B} / B^2$ perpendicular to the magnetic field, but each particle has its own velocity V along $\hat{\mathbf{b}} = \mathbf{B} / B$, satisfying

$$\frac{dV}{dt} = \frac{e}{m} E_{\parallel} + \hat{\mathbf{b}} \cdot \nabla \left(\frac{U^2}{2} - \mu B \right) + V \mathbf{U} \cdot (\hat{\mathbf{b}} \cdot \nabla \hat{\mathbf{b}}). \quad (16)$$

Here, e and m denote the charge and mass of the particle, and μ is its magnetic moment, $v_{\perp}^2 / B$. The charge to mass ratio e/m is regarded as a large parameter, and $E_{\parallel}$ is neglected in Faraday's law. The parallel electric field is determined self-consistently by demanding that the electrical current computed from (16) vanish in order e/m , i.e., $j_{\parallel} = \sum (e_l / m_l) \rho_l \langle V_l \rangle = 0$, where l denotes the different species. Because of the large parameter e/m , this does not imply that $\mathbf{B} \cdot \nabla \times \mathbf{B}$ is zero, only that it is $O[(e/m)^0]$.

The kinetic equation for the linearized Eq. (16) can be written, after some algebra, as

$$\begin{aligned} & \left(\delta f - \frac{\delta B}{B} f_0 + \frac{\kappa}{k_{\parallel}} \frac{\partial f_0}{\partial V} \right) - \left[\frac{e}{m} E_{\parallel} + i k_{\parallel} \left(\frac{X^2}{r} \xi_r - \mu \delta B \right) \right] \\ & \times \frac{i \partial f_0 / \partial V}{\tilde{\omega} + k_{\parallel} V_s} = 0, \end{aligned} \quad (17)$$

where $V_s = V - V_0$, $V_0 = \mathbf{v} \cdot \hat{\mathbf{b}}_0$ is the equilibrium mass velocity along the field lines, and $\kappa = [\mathbf{U} \cdot (\mathbf{B} \cdot \nabla \mathbf{B})]_{\parallel} / B^2$. The parallel motion is driven primarily by the force on a magnetic dipole and a fluid force that originates from equilibrium flow perpendicular to the magnetic field. The curvature term drops out when $\delta\rho$ and $\delta p_{\perp}$ are calculated:

$$\left(\delta\rho, \delta p_{\perp} - p_{\perp} \frac{\delta B}{B} \right) = \sum_l \int (1, \mu B) \delta f_l d\mu dV.$$

The parallel electric field is eliminated by means of the parallel current constraint, $j_{\parallel} = O[(e/m)^0]$, evaluated by multiplying (17) by e_l / m_l , and integrating

$$E_{\parallel} \sum_l \left(\frac{e_l}{m_l} \right)^2 J_{l,0} = i k_{\parallel} \sum_l \left(\frac{e_l}{m_l} \right) \left(\delta B J_{l,1} - \frac{X^2}{r} \xi_r J_{l,0} \right). \quad (18)$$

Here,

$$J_{l,n} = \int \frac{\mu^n}{V_s - V_p} \frac{\partial f_{0l}}{\partial V} d\mu dV, \quad (19)$$

and

$$V_p = -\tilde{\omega} / k_{\parallel} \quad (20)$$

is the parallel phase velocity in the frame moving with the plasma. After elimination of $E_{\parallel}$, the perturbations in perpendicular pressure and plasma density can be expressed in terms of the perturbations in magnetic field strength and in a "centrifugal potential" connected with the plasma flow:

$$\begin{aligned} & \delta p_{\perp} - \frac{2 \delta B}{B} p_{\perp} = a_1 B \delta B - a_2 \frac{\rho X^2}{r} \xi_r, \\ & \delta\rho - \frac{\delta B}{B} \rho = a_3 \rho \frac{\delta B}{B} - a_4 \frac{\rho X^2}{r} \xi_r. \end{aligned} \quad (21)$$

The coefficients are given by

$$\begin{aligned} & a_1 = I_{2,0} - I_{1,1} I_{1,1} / I_{0,2}, \\ & a_2 = a_3 = B \rho^{-1} (I_{1,0} - I_{1,1} I_{0,1} / I_{0,2}), \\ & a_4 = \rho^{-1} (I_{0,0} - I_{0,1} I_{0,1} / I_{0,2}), \end{aligned} \quad (22)$$

with

$$I_{n,m} = \sum_l \left(\frac{e_l}{m_l} \right)^m J_{l,n}. \quad (23)$$

Our expressions (21) for the plasma response are very simi-

lar in form to the expressions of Dobrott and Greene¹⁶ for $\nabla p_{\perp}$ and ∇p in the axisymmetric equilibrium problem.

In Appendix A we discuss the question of energy conservation in the linearized GCP system and show that this system has an energy integral when the kinetic damping vanishes. In Appendix B, we derive the coefficients a_1 through a_4 for fluid models with an equation of state. For these models, the condition for energy conservation can be formulated as a condition on the partial derivatives of $p_{\parallel}$ and $p_{\perp}$ with respect to ρ and B .¹⁹ This condition is, of course, satisfied by both MHD and the double adiabatic approximation.

C. Plasma response in GCP, MHD, and CGL

For a plasma in local thermodynamic equilibrium ($p_{\parallel} = p_{\perp} = p$, $p_e = p_i = p/2$, and Maxwellian distributions), the coefficients (22) are given by

$$a_1 = -\frac{p}{B^2} \left(\chi_e + \chi_i - \frac{(\chi_e - \chi_i)^2}{2(\chi_e + \chi_i)} \right),$$

$$a_2 = -\frac{2\chi_e\chi_i}{\chi_e + \chi_i}, \quad a_4 = \frac{\rho}{p} a_2, \quad (24a)$$

where

$$\chi = 1 + \xi Z(\xi), \quad \xi = -(\tilde{\omega}/|k_{\parallel}| \sqrt{2}v_{th}), \quad (24b)$$

and Z is the Fried-Conte plasma dispersion function.²¹ The form (24) for the coefficients is assumed for the numerical examples throughout this paper and leads to strong kinetic damping of the ion sound and slow magnetoacoustic waves. Sound waves of a hydrodynamic character can, of course, be recovered in a nonisothermal plasma with $T_e \gg T_i$.

It is of interest to compare the plasma responses of the various models GCP [(24)], MHD [$p_{\parallel} = p_{\perp} = p$ and $\delta(p/p^{\gamma}) = 0$], CGL [$\delta(p_{\perp}/\rho B) = 0$ and $\delta(p_{\parallel} B^2/\rho^3) = 0$], and MHD with a large parallel viscosity. The coefficients in the limit of small phase velocity, $|V_p| \ll |v_{thi}|$, are given in Table I. Here, it has been assumed that the electron and ion temperatures are equal, but not isotropic, and the distributions have been taken as bi-Maxwellians. For the viscous MHD coefficients, the parallel viscosity has been assumed to be infinite, i.e., we are concerned with the response on time scales much shorter than the collision time. Table I shows that the CGL equations reproduce the same type of dependence on $p_{\perp}/p_{\parallel}$ as the guiding center model; however, the numerical coefficients are incorrect. This is not surprising, since the CGL equations ignore the heat flux along the field lines and only apply in the limit of large phase velocity. It is

more surprising and of more significance that, for isotropic equilibria, MHD with the adiabatic index γ set to unity *exactly* reproduces the GCP coefficients when the parallel phase velocity is zero. It follows that for *static* equilibria (with finite shear) MHD, but not CGL, gives the same marginal point as GCP. The reason for the inaccuracy of the double-adiabatic approximation is the neglect of the heat flow along the magnetic field, which means that the temperatures cannot equilibrate along the field lines, contrary to the case of zero phase velocity in GCP and MHD with $\gamma = 1$. (It may be noted that, for *static* equilibria, only the value of a_1 is significant at marginal stability, and a_2 , a_4 , and, hence, γ , are immaterial. However, for special types of flow, such as rigid rotation, where marginal stability also occurs at $V_p = 0$, the coefficients related to compressibility a_2 and a_4 do influence the marginal point; see Appendix C.)

In general, when the equilibrium has mass flows, the parallel phase velocity is nonzero. In the inviscid fluid models, the coefficients a_1 through a_4 have singularities when the phase velocity equals the sound speed. These singularities are smeared out in GCP, and the GCP coefficients have large imaginary parts due to wave-particle resonance for parallel phase velocities around the thermal speed, which introduce damping in the macroscopic system. To emphasize the difference between GCP, MHD, and CGL, Fig. 1 shows $1 + a_2(V_p)$, which gives the density response to a perturbation in magnetic field strength. We note (a) the unphysical resonances and the absence of an imaginary part in the fluid models, (b) that CGL deviates from GCP already in the static case $V_p = 0$, and (c) that for small but finite phase velocity, the MHD coefficient not only ignores the dominant imaginary part, but also gives the sign opposite to that at GCP for the real part. It is obvious that fluid models can give spurious results for equilibria with mass flows, in particular, when the phase velocity of a mode is near the ion thermal speed.

D. The radial eigenvalue problem

Given the expressions (21) for $\delta\rho$ and $\delta p_{\perp}$, we can now close the system (11)–(13) and derive a pair of first-order

TABLE I. Values of the coefficients a_1 , a_2 , and a_4 , characterizing the parallel motion of the plasma, in the static limit $V_p = 0$, for GCP, CGL, and MHD, including the case of infinite parallel viscosity.

	$-a_1$	$-a_2$	$-a_4$
GCP	$2p_{\perp}^2/p_{\parallel} B^2$	$p_{\perp}/p_{\parallel}$	$\rho/p_{\parallel}$
MHD	$2p/B^2$	1	$\rho/\gamma p$
CGL	$p_{\perp}^2/3p_{\parallel} B^2$	$p_{\perp}/3p_{\parallel}$	$\rho/3p_{\parallel}$
MHD ($\mu_{\parallel} = \infty$)	$(2 - 9\gamma/4)p/B^2$	$-1/2$	0

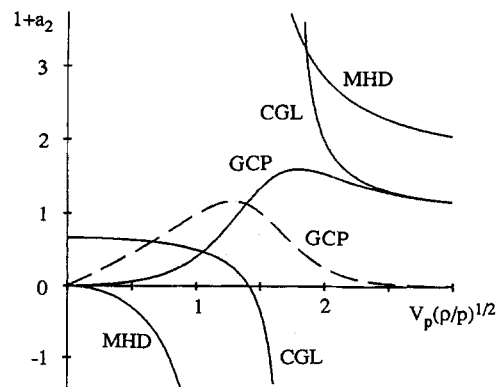


FIG. 1. Density response to perturbations in magnetic field strength, $1 + a_2$, as a function of the parallel phase velocity for different models of the parallel dynamics: GCP, MHD, and CGL. The solid lines give the real parts and the dashed lines the imaginary parts ($= 0$ for the fluid models).

equations for ξ_r and p_* . We first substitute the expression (21) for $\delta p_\perp$ into the definition of p_* and obtain

$$p_* = KB \delta B + H \xi_r / r, \quad (25)$$

where

$$K = 1 + 2p_\perp / B^2 + a_1, \quad (26)$$

$$H = \sigma B_\theta^2 - \rho v_\theta^2 - a_2 \rho X^2.$$

According to (25), K may be interpreted as the perturbation in total perpendicular pressure $p_\perp + B^2/2$ due to a unit perturbation in magnetic pressure $B^2/2$. An equation for $d\xi_r/dr$ is obtained directly from (25) using (9a) for δB and (12) for $\xi_\perp$, and dp_*/dr is computed from the radial equation of motion (11) [substituting (15), (21), and (25)]. Thus we finally obtain a radial eigenvalue problem of the same form as the MHD system derived by Appert *et al.*¹⁷

$$\frac{1}{r} \frac{d}{dr} (r \xi_r) = C_1 \xi_r - C_2 p_*,$$

$$\frac{dp_*}{dr} = C_3 \xi_r - C_4 p_*. \quad (27)$$

The coefficients in (27) are given by

$$C_1 = \frac{1}{r B^2} \left(-\frac{2TB_z E}{A} + B_\theta^2 + \frac{H}{K} \right), \quad (28a)$$

$$C_2 = \frac{1}{B^2} \left(-\frac{E^2}{A} + \frac{1}{K} \right), \quad (28b)$$

$$C_3 = A + r \frac{d}{dr} \left(\frac{\sigma B_\theta^2 - \rho v_\theta^2}{r^2} \right) - \frac{4B_z^2 T^2}{r^2 B^2 A} - a_4 \frac{\rho X^4}{r^2} + \frac{1}{r^2 B^2} \left(\frac{H^2}{K} + \frac{B_\theta^3}{F^2} (B_\theta A + 4FT) \right). \quad (28c)$$

Equations (26)–(28) give the MHD,⁷ viscous MHD,¹³ CGL, and GCP¹⁸ eigenvalue problems by substitution of the appropriate coefficients a_1 , a_2 , and a_4 from (22), (B2), or (B5). The system is manifestly Hamiltonian in $r\xi_r$ and p_* . This property would be lost if the symmetry $a_2 = a_3$ (22) [for the fluid models, condition (B1)] was violated. The Hamiltonian character of (26)–(28) has certain mathematical advantages, such as allowing a strong variational form; see Sec. IV A.

The information concerning the parallel dynamics is contained in K , H , and a_4 , and, in the fluid models, these show unphysical resonances at the sound speed. These resonances do not lead to singularity of the radial equation, because of interrelations between the coefficients (B2). However, true singularities of the radial eigenvalue problem occur at the zeros of K , corresponding to the slow wave continua. In the inviscid fluid models, the zeros of K occur for real frequencies. For instance,

$$K^{\text{MHD}} = 1 + \beta V_p^2 / (V_p^2 - v_s^2),$$

where $v_s^2 = \gamma p / \rho$ and $\beta = \gamma p / B^2$, has zeros at $V_p^2 = v_s^2 / (1 + \beta)$ close to its poles at $V_p^2 = v_s^2$. The important difference between the fluid models and GCP is that with the kinetic description of the parallel motion, the poles of a_1 and the zeros of K lie in the stable half-plane, if the distribution function is locally stable. Hence, for real ω , K^{GCP} is uniformly $1 + O(\beta)$.

In Sec. V, we discuss the problem of equilibrium in two and three dimensions and show that the question of ellipticity for the equilibrium equation can be answered by inspection of the radial eigenvalue problem (27). The first hyperbolic region in MHD occurs when K is negative and near zero. This hyperbolic region is completely removed in GCP by the kinetic treatment of the parallel dynamics.

The slow waves behave very differently in GCP and fluid theories. However, the Alfvén continuum remains essentially unchanged; $A = \rho \tilde{\omega} - \sigma F^2$ contains no kinetic integral, nor is it affected by parallel viscosity. Thus the continuum frequencies are real, and at an Alfvén resonance $r = r_A$ (where $A = 0$ and $F \neq 0$), ξ_r and p_* are both proportional to $\log(r - r_A)$, independent of the model used for the parallel dynamics. However, the discrete Alfvén eigenmodes become weakly damped in GCP.

III. LOCAL STABILITY

A. The Suydam index

In MHD, the stability of modes localized at the resonant surfaces can be analyzed simply from an indicial equation. To compute the characteristic exponents, we expand (27)–(28) around the resonant surface, $r = r_0$, where $F = 0$, with $\tilde{\omega}(r_0) = 0$, and set

$$\xi_r = \xi_0 x^\nu, \quad p_* = p_0 x^{\nu+1}, \quad x = r - r_0.$$

The characteristic exponents ν can be written in terms of the Suydam index D_0 as

$$\nu = -\frac{1}{2} \pm (1 - D_0)^{1/2}.$$

When D_0 is real, as in MHD or CGL, $D_0 > \frac{1}{4}$ implies the existence of an infinite sequence of unstable eigenmodes,¹⁴ and $D_0 < \frac{1}{4}$ is a necessary condition for stability, as shown by Suydam.²² For the generalized system (27)–(28),

$$D_0 = \frac{1}{\sigma - M^2} \left(\frac{q}{q' B_z} \right)^2 \left\{ -2MV \frac{B_z^2}{r B_\theta} \frac{q'}{q} - \frac{B^2}{B_\theta^2} \left[r \left(\frac{V^2}{r^2} \right)' + \frac{a_4}{\rho r^2} (V - MB_\theta)^4 \right] + \frac{1}{r^2} \left(-r(p_\parallel' + p_\perp') + \frac{H^2}{KB_\theta^2} - (\sigma + M^2) B_\theta^2 + 2(V - MB_\theta)^2 \right) \right\}, \quad (29)$$

where prime denotes differentiation with respect to r . In Eq. (29),

$$M = \frac{\sqrt{\rho} \tilde{\omega}'}{F'} = \frac{\sqrt{\rho} d/dr(\mathbf{k} \cdot \mathbf{v})}{d/dr(\mathbf{k} \cdot \mathbf{B})} \quad (30)$$

is the differential Mach number. We have also introduced

$$V = \sqrt{\rho} v_\theta, \quad (31)$$

and, to connect with the theory of toroidal devices, the safety factor $q = r B_z / R B_\theta$, where R is an assumed major radius. The phase velocity

$$V_p = -M v_A, \quad (32)$$

entering the arguments of the Z functions in (29), is real. The MHD expression⁷ for D_0 can be recovered by substitut-

ing Eq. (B2) for a_1 through a_4 . In Appendix C, we compare the Suydam criteria of the three models in the case of non-sheared flow, $M = 0$. As is well known,²³ the CGL condition is more optimistic than MHD, but MHD with $\gamma = 1$ reproduces the isotropic GCP condition *exactly* in the absence of sheared flow. It is also shown in Appendix C that the Suydam criterion obtained from MHD with infinite parallel viscosity is even more optimistic than the CGL criterion.

When the flow has shear, the guiding center result for the characteristic exponent differs from its fluid counterparts in two essential ways. First, in MHD, D_0 has a singularity due to resonance with the slow magnetoacoustic waves⁷:

$$D_{\text{res}}^{\text{MHD}} = \left(\frac{q}{q' B_z} \right)^2 \frac{[(V - MB_\theta)^2 + M^2(B_\theta^2 - V^2)]^2}{(1 - M^2)r^2 B_\theta^2} \times \frac{1 + \beta}{\beta - M^2(1 + \beta)}. \quad (33)$$

In GCP, this resonance is broadened kinetically. The slow wave continuum still appears through the terms $\propto K^{-1}$ in C_1 to C_3 , but the zeros of K now correspond to oscillations that are damped in time. Since $K^{-1} = 1 + O(\beta)$ for all real phase velocities in GCP, we can assert that the term in D_0^{GCP} representing the interaction with the slow magnetoacoustic wave is small in a low-beta plasma with sub-Alfvénic flows. The same conclusion could have been drawn with less effort if we had crudely added an imaginary part of order beta to the resonant denominator, $\beta - M^2(1 + \beta)$, of $D_{\text{res}}^{\text{MHD}}$ in Eq. (33).

The other modification, perhaps of more theoretical interest, is that the Suydam index in GCP becomes complex whenever M is nonzero. As a consequence, the infinite sequences of unstable eigenmodes occurring in MHD when $D_0 > \frac{1}{4}$ no longer exist. An asymptotic analysis similar to that in Ref. 7 now predicts that the growth rates lie on spirals in the complex plane, $\Gamma_n = i\omega_n(r_0) = \Gamma_0 \exp(-n\pi/s)$, $n = 1, 2, \dots$, with $s = (D_0 - \frac{1}{4})^{1/2}$ complex. The spiral will eventually cross the real ω axis and the higher modes will be damped. (When the sequence reappears on the unstable side, it is on an unphysical Riemann sheet, corresponding to an integration path in the complex r plane that cannot be deformed to the real axis, and so must be discarded.) Thus, when D_0 is complex, no truly asymptotic statement can be made concerning the existence of unstable modes. Evidently, the complex Suydam parameter of GCP does not have as clear an interpretation as its real-valued MHD counterpart.

B. Discrete slow modes

The MHD Suydam criterion predicts unconditional instability when M^2 approaches $\beta/(1 + \beta)$ from below. As we have shown, this loss of stability does not occur in GCP. MHD theory⁷ also predicts that, as M^2 is increased just above $\beta/(1 + \beta)$, the subsonic instabilities transform into infinite sequences of unstable discrete slow modes, accumulating at the edge of the slow wave continuum, where K has a quadratic zero in r .

In the guiding center description, the slow wave continuum, $K = 0$, lies in the stable half-plane (except at $k_{\parallel} = 0$,

where it touches the real axis) and infinite sequences of unstable slow modes cannot exist, for several reasons. First, similar to the case for the modes localized at the resonant surface $k_{\parallel} = 0$, the D parameter for the slow modes, which determines the characteristic exponent, is complex in GCP. Consequently, the unstable sequences can at most be finite. Second, if an infinite sequence did exist, its accumulation frequency would lie in the stable half-plane. We expect that, in most cases, even the low order discrete slow modes that are potentially unstable in MHD⁷ are stable in GCP, if they exist at all. (Of course, if a locally unstable particle distribution is chosen, K can have zeros in the unstable half-plane, giving rise to an unstable slow wave continuum.) Third, as K^{GCP} is complex, the equilibrium profiles for which it has a quadratic zero for any ω , are a set of measure zero. Stated differently, the complex slow wave continua in GCP do not have "edges."

We conclude that the guiding center spectrum is genuinely different from that of MHD with respect to anything related to the slow waves, and for sheared flows with non-zero M these differences have important consequences for stability.

C. Numerical results

To see numerically the effects of flows on the local modes, we have computed the growth rate of the most unstable mode for an RFP-type equilibrium varying the pressure and axial flow. The equilibrium has minor radius $a = 1$, isotropic pressure, and

$$\begin{aligned} B_\theta &= J_1(\lambda r), \quad B_z = J_0(\lambda r) [1 - 2P_0\sigma(r_0 - r)]^{1/2}, \\ \rho &= 1 - 0.9r^2, \quad p = P_0 J_0^2(\lambda r) \sigma(r_0 - r), \\ v_\theta &= 0, \quad v_z = v_0(1 - r^2), \\ \lambda r_0 &= 2.40483 \text{ (first zero of } J_0), \quad \lambda = 2.8, \end{aligned} \quad (34)$$

where σ now denotes the Heaviside step function. The mode has $m = 1$ and $k = 2.4$. The result is shown as plots of constant growth rate in Figs. 2(a) (GCP) and 2(b) (MHD).

As noted in Ref. 7, MHD gives very pessimistic predictions for marginal stability when the flow is about sonic. However, Fig. 2(b) shows that the growth rate of the flow driven instability can be extremely small and clearly outside the region where ideal MHD is reliable. By contrast, GCP predicts that the flow somewhat increases the pressure gradient stable to local modes. It also makes the marginal stability boundary more sharply defined at finite flow speed, as is seen from the piling up of the growth rate contours with increasing v_z in Fig. 2(a). The main difference between the stability diagrams [2(a) and 2(b)] is that the region around the sound speed, which is *weakly unstable* even for small pressure gradients in MHD, becomes *stable* in the guiding center description. However, also at a more robust growth rate, 10^{-2} , GCP and MHD give opposite results with respect to the influence of flow on the allowable pressure gradient. Thus, in GCP, sheared flow can be favorable for local stability, due, in part, to the effect of kinetic damping.

D. Large aspect ratio limit

For toroidal devices, the Suydam criterion must be replaced by the Mercier criterion, which still remains to be

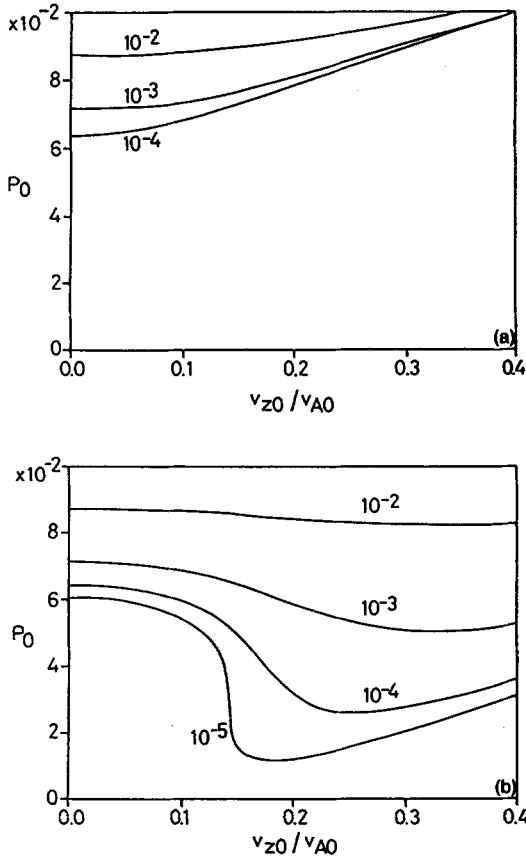


FIG. 2. Contours of equal growth-rate in GCP (a) and MHD (b) for the resonant mode $m = 1$, $k = 2.4$ on the equilibrium (34) with $\lambda = 2.8$. The MHD description gives rise to a large region of weak instability for near-sonic flows. This region is completely stable in GCP.

worked out in the presence of mass flow. However, certain information relevant to tokamaks can be obtained from the Suydam criterion (29) by considering a large aspect ratio ordering, $\beta = O(\epsilon^2)$, $B_\theta/B_z = O(\epsilon)$. It is seen directly from the GCP expression for D_0 that sonic flows [i.e. we assume that $M, V/B_\theta = O(\epsilon)$] only give $O(\epsilon^2)$ contributions to D_0 , and for tokamaks with $q \gtrsim 1$, these must be discarded because of the ordering. (Notably, the same conclusion cannot be drawn in the inviscid fluid models, because of the unphysical resonance with the slow magnetoacoustic wave.) However, for Alfvénic flows [$M, V/B_\theta = O(1)$], D_0 will be order unity also in GCP and, consequently, relevant information can be extracted from Eq. (29). [It must be remembered, however, that the assumed ordering implies $\rho v_z^2/B_z^2 = O(1)$, and that, in a torus, such flows can cause ballooning instability because of the toroidal curvature. Moreover, it is necessary to assume $\beta = O(\epsilon)$ to avoid equilibria with pathological density distribution within the magnetic surfaces.] In the lowest order, only the three first terms in (29) contribute

$$D_0^{(0)} = \frac{1}{1 - M^2} \left(\frac{q}{rq'B_\theta} \right)^2 \left[-2MVB_\theta \frac{rq'}{q} - r^3 \left(\frac{V^2}{r^2} \right)' - \frac{a_4}{\rho} (V - MB_\theta)^4 \right]. \quad (35a)$$

The second term in square brackets is essentially that which determines the stability of an unmagnetized fluid between two rotating cylinders, where the angular frequency v_θ/r must increase outward for ideal stability.²⁴ From the properties of the Z function [or by using the fluid equivalent (B2)], we see that $a_4 \approx -\rho/p$ for $M^2 \ll \beta$ and $a_4 \sim \rho/B^2 M^2$ when $M^2 \gg \beta$. Thus the term involving a_4 can be discarded by order of magnitude, except when $M^2 < \beta$ and the cross-field flow $V - MB_\theta$ is $O(\epsilon)$. In this case, the real part of a_4 is negative, and, hence, the a_4 term is destabilizing. (In the opposite case, $M^2 \gg \beta$, the term is stabilizing but “small.”) If β is ordered as ϵ , the a_4 term can always be neglected to lowest order, and (35a) can be written in a more transparent form

$$D_0^{(0)} = \frac{-(q/q'B_\theta)^2}{r(1 - M^2)} \left(\rho' v_\theta^2 + \frac{2B_\theta}{B_z} \rho v_\theta v_z' \right). \quad (35b)$$

To be more explicit, we now assume that B_θ , B_z , and v_z are positive and the flow profile is monotonic with v_z vanishing at the edge, so that $v_z' < 0$. According to (35b), this implies that azimuthal flow in the range $-2B_\theta \rho v_z' / \rho' B_z < v_\theta < 0$ is stabilizing when $\rho' < 0$. Thus the flow must be Alfvénic and have a pitch that is *opposite* to that of the magnetic field to give an $O(1)$ stabilizing contribution to D_0 . It is clear from (35b) that the normal inward density gradient is destabilizing in the presence of azimuthal flow.

IV. GLOBAL MODES

A. Boundary conditions and energy integral

For free boundary modes, the solution of (27) must be matched to a vacuum (or cold plasma) solution with $p_1 = p_\parallel = 0$, and where the perturbed magnetic field can be expressed as the gradient of a scalar potential, $\mathbf{Q} = \nabla\phi$. The boundary condition for (27) is continuity of Q_r and $\delta p_1 + B \delta B$ at the plasma edge (limiter) at radius $r = a$, giving

$$p_* = \xi_a \left(\frac{1}{a} [(\sigma B_\theta^2 - \rho v_\theta^2)_p - (B_\theta^2 - \rho v_\theta^2)_v] - F_p F_v \frac{\phi}{\partial\phi/\partial r} \right)_a, \quad (36)$$

where subscripts p and v refer to plasma and vacuum, respectively. Equation (36) allows for discontinuities at the surface and also for finite flow of the edge plasma. Integrating $d/dr(p_* r \xi_r)$ from the center to $r = a$ using (27) and (36), we obtain

$$\begin{aligned} & \xi_a^2 \left([(\sigma B_\theta^2 - \rho v_\theta^2)_p - (B_\theta^2 - \rho v_\theta^2)_v] \right. \\ & \quad \left. - F_p F_v \frac{a\phi}{\partial\phi/\partial r} - \frac{(aC_1 - 1)}{C_2} \right)_a \\ & = - \int_0^a [f \xi'^2 + g \xi^2] dr, \end{aligned} \quad (37a)$$

with

$$f = \frac{r}{C_2}, \quad g = r^2 \left(\frac{rC_1 - 1}{r^2 C_2} \right)' - rC_3 + \frac{rC_1^2}{C_2} - \frac{1}{rC_2}. \quad (37b)$$

Equations (37) corresponds to the energy integral δW in the static case. The second-order radial equation, corresponding to the Hain-Lüst equation²⁵ in the static case, reads

$$(f\xi')' - g\xi = 0. \quad (38)$$

We note that this equation and the boundary condition (36) are variational derivatives of (37) with respect to ξ .

As the system with flow is nonself-adjoint, the integral (37) is not as useful as δW for the static problem. However, for a certain class of modes, namely, those locked to resistive walls, Eq. (37) gives valuable information. Such external modes, whose magnetic field has time to diffuse through a resistive wall, have growth times of the order of the resistive decay time $\tau_w (= L/R)$ of the wall, and this is typically much longer than the Alfvén time. Therefore, the frequency of a wall-locked nonresonant mode (i.e., one that does not have a resonant surface in the plasma) can be considered as zero inside the plasma. Thus Eq. (37) with $\omega = 0$ in f and g may be regarded as an expression for $\phi(\partial\phi/\partial r)^{-1}$ that determines the growth rate of the resistive wall modes by controlling Δ' at the wall.²⁶

B. Tokamak ordering

We now apply the analysis of Sec. IV A in the large aspect ratio, low-beta tokamak ordering already discussed for the Suydam criterion, assuming $\beta = O(\epsilon^2)$, $B_\theta/B_z = O(\epsilon)$, $\rho v_\theta^2/B_z^2 = O(1)$, $\rho v_z^2/B_z^2 = O(1)$. For simplicity, we once again discard the term proportional to a_4 , which is correct if the flow is supersonic. (If this term were kept, it would be destabilizing for subsonic flows, as in the case of local modes.) Within the large aspect ratio ordering, the vacuum field is represented in terms of a vector potential with only an axial component Ψ , i.e., $\mathbf{Q} = \nabla\Psi \times \hat{z}$. The logarithmic derivatives of the scalar and vector potentials are related by $\phi/\phi' = rQ_\theta/imQ_r = (r^2/m^2)\Psi'/\Psi$. For equilibria that are continuous at the surface $r = a$, the long-thin approximation of (37) with $B_z(r) \approx \text{constant}$ becomes

$$\begin{aligned} & \xi_a^2 [a^3 F^2 \Delta_p + B_\theta^2 (m^2 - n^2 q^2) \\ & - \rho v_\theta^2 m^2 + \rho a^2 (\omega - kv_z)^2]_a \\ & = \int_0^a (f\xi'^2 + g\xi^2) dr = \delta W_p, \end{aligned} \quad (39a)$$

where

$$\begin{aligned} f &= r^3 (F^2 - \rho \tilde{\omega}^2), \\ g &= r(m^2 - 1)(F^2 - \rho \tilde{\omega}^2) + r^2 \frac{d}{dr} [\rho (\omega - kv_z)^2], \end{aligned} \quad (39b)$$

$$F = (B_\theta/r)(m - nq), \quad \Delta_p = \Psi'/\Psi|_{r=a}.$$

Equations (39) give the straight tokamak,^{27,28} or lowest-order reduced-MHD²⁹ ordering of the linear stability problem for equilibria with Alfvénic flows.

For the wall-locked modes, we must set $\omega = 0$ in f and g and match the logarithmic derivative of Ψ obtained from (39) to the solution in the vacuum region and the resistive wall. If the wall is modeled as a thin finitely conducting shell at $r = b$, the logarithmic derivative Δ_v of the external solution for Ψ depends on the frequency according to³⁰

$$\begin{aligned} \Delta_v(\omega) &= \frac{\Delta_v(0) + p\Delta_v(\infty)}{1 + p}, \quad p = \frac{i\omega\tau_w}{|m|} \left[1 - \left(\frac{a}{b} \right)^{2|m|} \right], \\ \Delta_v(0) &= -\frac{|m|}{a}, \quad \Delta_v(\infty) = -\frac{|m|}{a} \frac{1 + (a/b)^{2|m|}}{1 - (a/b)^{2|m|}}. \end{aligned} \quad (40)$$

The growth rate is obtained as the solution of $\Delta_v(\omega) = \Delta_p$. The expression given in (40) for $\Delta_v(\omega)$ shows that, as long as $\Delta_v(0) > \Delta_p > \Delta_v(\infty)$ (i.e., if the mode is stable with an ideally conducting wall and unstable in the absence of a wall), the growth rate of a resistive wall mode increases when Δ_p decreases. As a consequence, flow inside the plasma is destabilizing for two reasons, which can be seen from Eqs. (39). First, the effect of the field bending term F^2 is reduced by the inertial contribution $\rho \tilde{\omega}^2$. (In the present discussion we assume that $\rho \tilde{\omega}^2 < F^2$; otherwise, it would be necessary to consider coupling to the continua.⁷) Furthermore, if the axial flow speed decreases toward the edge, the contribution $r^2 k^2 (\rho v_z^2)'$ to g is also negative, i.e., destabilizing. The only possibility of stabilization by mass flow is if the flow speed is finite at the edge. It is clear from (38) that finite azimuthal flow at the edge is stabilizing. Unfortunately, such profiles appear difficult to achieve in practice.

C. Numerical results

In this section, we give numerical results for global modes obtained from the full system (27) with the boundary condition (36). Figure 3 shows the effect on an $m = 2, n = 1$ external kink mode of adding a flow $v_z = v_{z0}(1 - r^2/a^2)$ to a tokamaklike equilibrium with $R/a = 10$, $q_a = 1.86$, and $q_0 = 1.05$. The wall is located at $r = b = 1.15a$ and its time constant is $\tau_w = 10^4 \tau_A$. The mode is unstable already in the absence of flow, and Fig. 3 shows that the growth rate is increased by the flow. The curves are labeled: (a) no azimuthal flow, (b) $v_\theta = v_z kr/m$, giving $\mathbf{k} \cdot \mathbf{v} = 0$, and (c) $v_\theta = -v_z kr/m$. Curve (b), for the case of vanishing Doppler shift, shows destabilization by the outward decrease in v_z^2 . Destabilization by inertia is also evident, since out of the three curves in Fig. 3, (c) is the most unstable case. In addition, curve (c) suggests that the instability changes character at a certain flow speed. This is more clearly seen from Fig. 4, where growth rates are shown for the same equi-

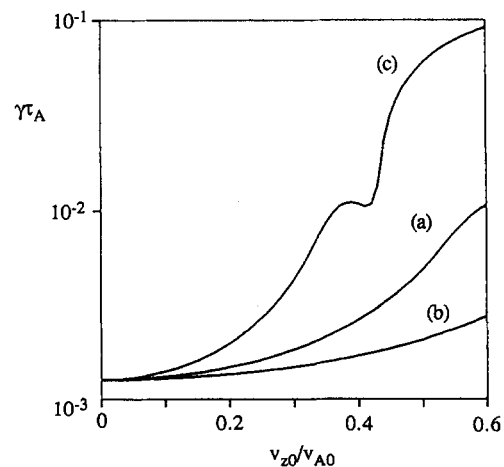


FIG. 3. Growth rates of the $m = 2, n = 1$ external kink mode versus flow speed for a tokamaklike equilibrium with $q_0 = 1.05$ and $q_a = 1.86$ and a resistive wall at $b = 1.15a$ with time constant $\tau_w = 10^4 \tau_A$. The curves are labeled (a) no azimuthal flow, (b) $v_\theta = kv_z/m$, giving $\mathbf{k} \cdot \mathbf{v} = 0$, and (c) $v_\theta = -kv_z/m$.

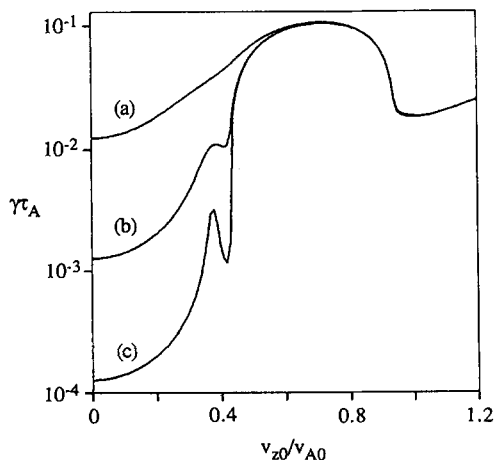


FIG. 4. Growth rates corresponding to case (c) of Fig. 3 for different wall time constants: (a) $\tau_w = 10^3 \tau_A$, (b) $\tau_w = 10^4 \tau_A$, and (c) $\tau_w = 10^5 \tau_A$. For $v_{z0} < 0.4 v_{A0}$ the mode is a resistive wall instability, whereas for $v_{z0} > 0.4 v_{A0}$, it is unstable with an ideal wall.

librium as in Fig. 3, but where the time constant of the wall is varied: (a) $\tau_w = 10^3 \tau_A$, (b) $\tau_w = 10^4 \tau_A$, and (c) $\tau_w = 10^5 \tau_A$. The transition at $v_{z0} \approx 0.4$ clearly corresponds to the onset of instability with an ideal wall. An interesting feature in Fig. 4 is that for this profile, the growth rate of the ideal mode shows a local minimum when the flow is about Alfvénic.

Figure 5 shows the growth rate of the cylindrical internal kink ($m = 1$) for a similar equilibrium with $q_0 = 0.95$ and $v_\theta = v_z kr/m$. Figure 6 shows the growth rate of an $m = 1$ external mode for the RFP equilibrium (34) with $P_0 = 0.03$ and purely axial flow. In this case, $k = -0.6$, the wall is located at $b = 1.1a$, and its time constant is $10^4 \tau_A$.

It is evident from Figs. 3–6 that mass flow increases the growth rates of the global modes. This is in agreement with the large aspect ratio theory for resistive wall modes in Sec. IV B. The numerical results also show that flow is destabiliz-

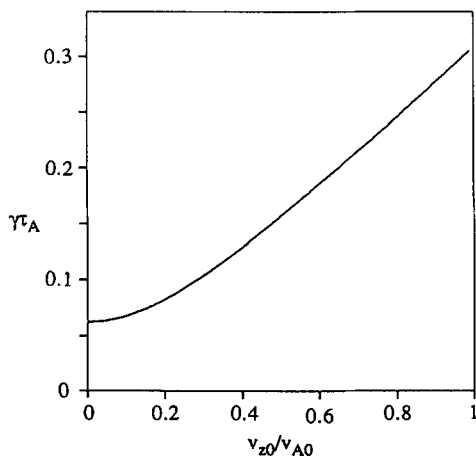


FIG. 5. Growth rate of the internal $m = 1$ kink mode for a tokamaklike equilibrium with $q_0 = 0.95$ and $v_\theta = kv_z/m$.

ing for the cylindrical internal kink and external modes in RFPs. The curves in Figs. 3–6 only show growth rates of modes that are unstable in the absence of flow, but we also find that flow tends to reduce the region of stability. Thus the marginal value of central q is decreased for the external kink (Fig. 3) and increased for the internal kink (Fig. 5) by the mass flow.

V. EQUILIBRIUM PROBLEM IN TWO AND THREE DIMENSIONS

The derivations presented in Sec. II show many similarities to the GCP calculation of toroidal equilibrium given by Dobrott and Greene.¹⁶ These authors found that the first hyperbolic region that occurs in the MHD equilibrium problem, when the poloidal flow is of the order of the poloidal sound speed,¹⁵ is eliminated when kinetic effects along the field lines are accounted for. In this section, we discuss some important consequences of the linear system (27) for equilibria in two and three dimensions.

First, we note that the hyperbolicity or ellipticity of the equilibrium problem depends only on the *local* properties of the linearized equilibrium equation. This equation is a special case of the linear eigenvalue problem for perturbations having zero frequency ω in the laboratory frame (however, the Doppler shifted frequency $\tilde{\omega} = \mathbf{k} \cdot \mathbf{v}$ is nonzero because of the flow). As the information required is local, we can use the cylindrical eigenvalue problem (27), take the limit of m and k large, and keep only the leading order terms

$$\begin{aligned} \frac{d\xi_r}{dr} &= \frac{1}{B^2} \left(\frac{E^2}{A} - \frac{1}{K} \right) p_*, \\ \frac{dp_*}{dr} &= A \xi_r. \end{aligned} \quad (41)$$

From (41), we obtain a radial wave number in the WKB sense,

$$k_r^2 = \frac{1}{B^2} \left(\frac{A}{K} - E^2 \right), \quad (42)$$

which is simply the dispersion relation for the slow and fast

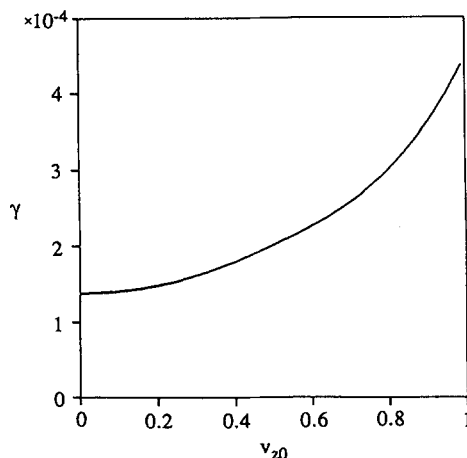


FIG. 6. Growth rate of the $m = 1$ external mode with $k = -0.6$ for the RFP equilibrium (34) with $\lambda = 2.8$, $P_0 = 0.03$, and $v_\theta = 0$. The resistive shell is placed at $r = b = 1.1a$ and has the time constant $\tau_w = 10^4 \tau_A$.

magnetoacoustic waves. (As usual, the Alfvén wave, $A = 0$, decouples from this dispersion relation. The r direction is singled out as that which is perpendicular to both $\mathbf{B}_0$ and $\mathbf{v}_0$.) In the following, we consider cases where the kinetic damping vanishes so that K , and hence k_r^2 , are real when m and k are real.

The distinction between the elliptic and hyperbolic regions only depends on the local properties of the equilibrium equation and can, therefore, be made from the local dispersion relation (42). When $\omega = 0$, K depends on m/r and k only through the ratio m/kr , and we see that the GCP equilibrium problem is nondispersive in the short wavelength limit, exactly like MHD and CGL. [In more precise terms, with $\kappa^2 = k_r^2 + m^2/r^2 + k^2 k_r/\kappa$ from (42) depends on $m/r\kappa$ and k/κ but not on κ .] If the dispersion relation (42) has solutions with k_r , k , and m/r nonzero and all *real*, the linear equation (41), when transformed into configuration space, has real characteristics, along which discontinuities may be propagated, i.e., the equation is of hyperbolic type. Thus the equilibrium problem is hyperbolic if k_r^2 in (42) is positive for *any* real m and k . On the other hand, if k_r^2 in (42) is negative for *all* real m and k with $k^2 + m^2/r^2 > 0$, real characteristics cannot exist. In this case, the equilibrium problem is elliptic and, therefore, well behaved.

For the moment, we consider explicitly the axisymmetric case and set $k = 0$ in (42). Then, it is easy to verify that the MHD equilibrium equation is elliptic when

$$\frac{(1 - M_p^2)(\beta - M_p^2)}{\beta - M_p^2(1 + \beta)} > -\frac{B_z^2}{B_\theta^2}, \quad (43)$$

where $\beta = \gamma p/B^2$ and $M_p = \sqrt{\rho} v_\theta/B_\theta$ is the poloidal Mach number. Equation (43) is the ellipticity condition for the MHD Grad-Shafranov equation given by Zehrfeldt and Greene.¹⁵ We note that MHD theory predicts a first hyperbolic region that is very narrow for tokamaks; the equilibrium equation is hyperbolic for M_p^2 between $\beta/(1 + \beta)$ and approximately $\beta/(1 + \beta B_z^2/B_\theta^2)$. The present derivation shows that the first hyperbolic region in MHD occurs because, in this narrow range of Doppler frequency due to the flow, perturbations that have zero frequency in the laboratory frame can propagate radially as slow MHD waves. This occurs when K is small and negative.

If the restriction to axisymmetry is abandoned, we must consider wave vectors $\mathbf{k}$ in the plane spanned by $\mathbf{B}$ and $\mathbf{v}$, making arbitrary angles θ_B and θ_v with respect to $\mathbf{B}$ and $\mathbf{v}$. In terms of the Alfvén Mach number $M_A = \sqrt{\rho} v/B$, or $\Phi = M_A^2 \cos^2 \theta_v$, the ellipticity condition in MHD can be written as

$$\frac{(\cos^2 \theta_B - \Phi)(\beta \cos^2 \theta_B - \Phi)}{\beta \cos^2 \theta_B - (1 + \beta)\Phi} > -\sin^2 \theta_B, \quad (44)$$

which must hold for all θ_B between 0 and $\pi/2$. Condition (44) will be violated when Φ is close to $\beta \cos^2 \theta_B/(1 + \beta)$, i.e., by resonance with the (unphysical) slow wave in MHD. Such a resonance will clearly occur for some direction of $\mathbf{k}$ when the flow is supersonic, $M_A^2 > \beta/(1 + \beta)$. However, Eq. (44) also shows that if θ_B is close to $\pi/2$, the ellipticity condition can be violated for arbitrarily small Φ and, hence,

for arbitrarily small flow (except if the flow is field aligned). Thus, discontinuities can be propagated in three-dimensional MHD equilibria by slow waves in a direction almost perpendicular to the magnetic field, even for arbitrarily small cross-field flow. Only field aligned flow gives a finite threshold for hyperbolicity in three-dimensional MHD:

$$(1 - M_A^2)(\beta - M_A^2)/[\beta - (1 + \beta)M_A^2] < 0. \quad (45)$$

Thus, for field-aligned flow, the three-dimensional MHD equilibrium equation is hyperbolic when $\beta/(1 + \beta) < M_A^2 < \beta$ or $M_A^2 > 1$.

We conclude that *the MHD system without restriction to axisymmetry is hyperbolic for any nonzero flow across the field lines*. Thus calculations of stellarator equilibria with mass flows requires GCP theory (or any other theory that eliminates the slow wave).

The guiding center result for equilibria with mass flow is entirely different in that $K \approx 1$ in a low-beta plasma, and slow magnetoacoustic waves do not exist, except in cases with pathological particle distributions. In Fig. 7, we plot $\text{Re}(k_r^2)$ as a function of M_p for MHD and GCP. Although the equilibrium has high beta, $p/B^2 = 0.1$, and small aspect ratio, $r/R = 0.3$, the first hyperbolic region is narrow in MHD. Figure 7 shows very clearly that this region is completely removed in GCP. We note that the absence of the first hyperbolic region in the GCP equilibrium problem¹⁶ is connected with the nonvanishing of the *real* and not the imaginary part of K since, in toroidal equilibria, the distribution functions must be symmetric in the trapped region (cf. Ref. 16, where the real-valued integral $C_\perp$ corresponds to our a_1). It is easy to see that if we make the low-beta approximation $K \equiv 1$, which is uniformly valid with respect to the flow speed in GCP, then $k_r^2 < 0$ for all $M_A^2 < 1$ even in the general three-dimensional case, i.e., *the GCP equilibrium equation remains elliptic as long as the flow is sub-Alfvénic*.

VI. CONCLUSION

The guiding center plasma model, which treats the motion of the particles along the magnetic field kinetically,

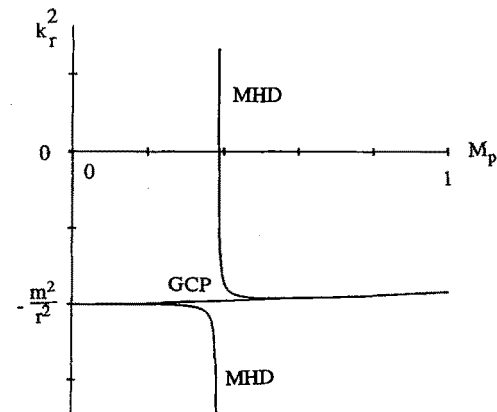


FIG. 7. Square of the radial wavenumber in the axisymmetric equilibrium problem versus poloidal Mach number $M_p = \sqrt{\rho} v_\theta/B_\theta$ in GCP and MHD for an equilibrium with $p/B^2 = 0.1$ and $r/R = 0.3$. The Grad-Shafranov equation is hyperbolic when $k_r^2 > 0$.

gives significantly different results than MHD when the equilibrium has sheared mass flow. For isothermal plasmas, GCP completely removes the loss of stability that occurs in MHD for near-sonic flows. In the present paper, we have applied GCP theory to investigate the ideal stability of cylindrical equilibria.

In the straight tokamak ordering, sonic flows have little influence on stability, and the flows must be Alfvénic to have a strong effect in GCP. The leading order contribution to the Suydam criterion is given by Eq. (35). Flows, whose axial component decays monotonically toward the edge, can be stabilizing for the local modes only if the flow has a pitch that is opposite to that of the magnetic field. We find numerically for high-beta RFP equilibria that flow can be stabilizing for the local modes, due, in part, to the kinetic damping of the slow waves.

For the external modes locked to resistive walls, we have shown analytically in the large aspect ratio tokamak limit that flow is always destabilizing if the flow speed vanishes at the plasma edge [see Eq. (39)]. Numerical evidence supports this conclusion also for RFP configurations. Despite numerous attempts, we have not been able to find any flow profile, where the speed vanishes at the edge, that improves the stability of the external modes. It is noteworthy, however, that stabilization can be obtained if the edge plasma ($r > a$) rotates in the poloidal direction.

Although we have not treated resistive instabilities in the present paper, it should be mentioned that global resistive modes can be stabilized by weak flows in devices with finitely conducting walls. As shown in Refs. 26 and 30, resistive walls can stabilize tearing modes if the Doppler shift due to plasma rotation is large compared with both resistive growth rates and the inverse of the wall time. Under these circumstances, the mode cannot diffuse through both the resistive layer and the wall, and, therefore, either the resonant surface or the wall must behave “ideally.” Thus, resistive modes can be stabilized by rotation if the plasma is surrounded by walls of finite conductivity. However, sheared flows can also destabilize tearing modes by influencing the dynamics inside the resistive layer.³¹

For the problem of toroidal equilibrium, Dobrott and Greene¹⁶ showed that the guiding center plasma behaves very differently than the MHD fluid, and that the hyperbolic region found in MHD for poloidal flows close to the poloidal sound speed is removed in GCP. This first hyperbolic region is a result of the propagation of slow MHD waves, which are practically eliminated in GCP. For three-dimensional equilibria, the difference between the kinetic and fluid models is even more dramatic; the MHD equilibrium equation becomes hyperbolic for arbitrarily small flows across the magnetic field, whereas the GCP equilibrium problem remains elliptic as long as the flow is sub-Alfvénic.

Our study has shown that a kinetic description of the parallel dynamics is needed to understand equilibrium and stability of plasma with mass flow. An inevitable, but unfortunate, consequence is that the two-dimensional stability problem in toroidal geometry becomes considerably more difficult for rotating plasmas than in the static case, as the kinetic equation is nonlocal along the field lines. (This is,

fortunately, easy to deal with in the cylindrical analysis.) It appears that an understanding of the stability for rotating plasmas, comparable to the detailed picture that has emerged for static equilibria, will present significant difficulties and remains a major challenge to plasma theory. The cylindrical results indicate that, although it is possible to improve on local stability by adding mass flow, depending on details of the profiles, flow is always destabilizing for external kink modes.

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APPENDIX A: ENERGY RELATIONS

Here, we discuss energy conservation in systems with anisotropic pressure, which is related to the symmetry $a_2 = a_3$ in Eq. (22). The linear equation of motion (1) has the form

$$\rho \frac{\partial^2 \xi}{\partial t^2} + 2\rho \mathbf{v} \cdot \nabla \frac{\partial \xi}{\partial t} = \mathbf{F}[\xi] + \mathbf{G}[\delta p_\perp, \delta p_\parallel]. \quad (\text{A1})$$

The energy balance equation is obtained by dotting (A1) with $\xi(\mathbf{r}, t)$ [or, if $\exp(i\omega t)$ dependence is assumed, by taking the real part of the scalar product with $-\omega^* \xi^*$] and integrating. The convection term on the left-hand side of (A1) is an antisymmetric operator acting on $\partial \xi / \partial t$, and does not affect the energy balance. For fluid models, where $\delta p_\perp$ and $\delta p_\parallel$ are expressed in terms of ξ by an equation of state, energy conservation is equivalent to the condition that $\mathbf{F} + \mathbf{G}$ is self-adjoint in terms of ξ . (Thus, if $\mathbf{v} = 0$, the *eigenvalue* problem for such a model is self-adjoint if the system is energy conserving, but self-adjointness is clearly lost when mass flow is added, whether $\mathbf{F} + \mathbf{G}$ is energy conserving or not.) A necessary condition for conservation of energy within the linear system (A1) is that $\mathbf{F} + \mathbf{G}$ be Hermitian for real ω .

In the GCP formalism, $\xi_\parallel$ cannot be treated as a virtual displacement on the same footing as ξ_r and $\xi_\perp$. Rather, $\xi_\parallel$ has to be determined, together with $\delta\rho$, $\delta p_\perp$, and $\delta p_\parallel$, from ξ_r and $\xi_\perp$ (or more conveniently, ξ_r and δB) at an assumed value of ω . We consider the system to be energy conserving if $\mathbf{F} + \mathbf{G}$ in (1) is Hermitian when the frequency is real. Dotting (5) with ξ^* to produce a quadratic form, we find one piece that is not automatically energy conserving, namely

$$\frac{\delta B^*}{B} \delta p_\perp + \left(\frac{\delta \rho^*}{\rho} - \frac{\delta B^*}{B} \right) (\delta p_\parallel - \Delta B \delta B). \quad (\text{A2})$$

The expression in Eq. (A2) is the sum of the parallel and perpendicular contributions to the product of plasma compression and pressure perturbation. By substituting $\delta p_\parallel$ from (15), and $\delta\rho$ and $\delta p_\perp$ from (21), we find that $\mathbf{F} + \mathbf{G}$ is Hermitian if $a_2 = a_3^*$ and a_1 and a_4 are real for real ω . The symmetry $a_2 = a_3$ in (22), therefore, means that the linear GCP system has an energy integral when the a coefficients are real, i.e., when the kinetic damping vanishes. Wave-par-

ticle resonance introduces damping in the macroscopic system, which corresponds to irreversible conversion of kinetic energy into heat. In the inviscid fluid models, where all the coefficients a_1 – a_4 are real, energy conservation becomes equivalent to $a_2 = a_3$. In Appendix B, we show that this can be formulated as a condition on the partial derivatives of the pressures with respect to ρ and B . The condition is, of course, satisfied by the MHD and CGL equations.

APPENDIX B: PARALLEL DYNAMICS IN MHD AND CGL

In this Appendix, we derive the coefficients a_1 – a_4 describing the parallel dynamics for fluid models with an equation of state $p_\perp = p_\perp(\rho, B)$, $p_\parallel = p_\parallel(\rho, B)$ and give the condition for energy conservation that corresponds to the relation $a_2 = a_3$ for GCP. By differentiating the equation of state, $\delta p_\parallel$ may be expressed in terms of $\delta\rho$ and δB , and the parallel equation of motion (15) then gives $\delta\rho$ in terms of δB and $\xi, \rho X^2/r$, and thus a_3 and a_4 . Then, a_1 and a_2 are obtained by differentiating $p_\perp = p_\perp(\rho, B)$. In the inviscid fluid models, all coefficients are real and the condition for energy conservation becomes $a_2 = a_3$, which can be written as a condition on the partial derivatives of the pressures (see, also, Ref. 19 on toroidal equilibrium):

$$B \frac{\partial(p_\parallel/\rho)}{\partial B} = \rho \frac{\partial[(p_\parallel - p_\perp)/\rho]}{\partial \rho}. \quad (\text{B1})$$

We note that both the MHD and double adiabatic equations satisfy this condition. Other models may be introduced that have the same property, for instance, the isothermal model discussed in Ref. 19. When condition (B1) is satisfied, the fluid models give the following coefficients for the plasma response in (21):

$$a_1 + \frac{2p_\perp}{B^2} = \frac{\partial p_\perp}{B \partial B} + \frac{\partial p_\perp}{\partial \rho} \frac{\rho}{B^2} (1 + a_2),$$

$$a_2 = \frac{\partial p_\perp}{\partial \rho} a_4, \quad a_4 = \frac{1}{V_p^2 - \partial p_\parallel / \partial \rho}. \quad (\text{B2})$$

Here, $V_p = -\tilde{\omega}/k_\parallel$ stands for the parallel phase velocity in the plasma frame.

The coefficients a_1 – a_4 can also be defined for MHD with parallel viscosity, which was applied in a recent calculation by Gimblett.¹³ This model is valid in the opposite limit than the collisionless guiding center model, namely, when the Doppler frequency is small compared with the ion–ion collision time. The equilibrium pressure is isotropic, but a small anisotropy of the perturbed pressure results from anisotropic compression,³²

$$\delta p_\perp = \delta p + \mu_\parallel s, \quad \delta p_\parallel = \delta p - 2\mu_\parallel s. \quad (\text{B3})$$

Here, $\mu_\parallel = 0.96 p \tau_{ii}$ is the parallel viscosity, and

$$s = \hat{b} \cdot \nabla (\mathbf{v} \cdot \hat{b}) - \mathbf{v} \cdot (\hat{b} \cdot \nabla \hat{b}) - \frac{1}{3} \nabla \cdot \mathbf{v}$$

$$= i\tilde{\omega} \left(\frac{\delta B}{B} - \frac{2}{3} \frac{\delta \rho}{\rho} \right), \quad (\text{B4})$$

where we used (3) and (9). It can be shown from the expressions obtained by Braginskii³² that, for a cylindrical equilibrium, the parallel viscosity does not contribute to the linear perturbation in the isotropic pressure p . Substituting (B3)–

(B4) with $\delta p = (\gamma p / \rho) \delta \rho$ into the parallel equation of motion (15), we obtain the coefficients a_1 – a_4 :

$$a_1 + 2p/B^2 = (B^2 S)^{-1} [\gamma p V_p^2 + i\tilde{\omega} \mu_\parallel (\frac{1}{3} V_p^2 - 3v_s^2)],$$

$$a_2 = S^{-1} (v_s^2 - \frac{2}{3} i\tilde{\omega} \mu_\parallel / \rho), \quad a_4 = S^{-1}, \quad (\text{B5})$$

$$S = V_p^2 - v_s^2 - \frac{2}{3} i\tilde{\omega} \mu_\parallel / \rho, \quad v_s^2 = \gamma p / \rho.$$

Thus the viscous MHD model also satisfies the symmetry $a_3 = a_2$, although both coefficients are complex and the model does not conserve energy. The parallel viscosity has broadened the resonance at the sound speed similar to the kinetic effect in GCP.

Since the collisionless models apply for frequencies larger than ν_{ii} , it is of interest to compare these with the predictions of viscous MHD in the region $|\tilde{\omega}| \tau_{ii} \gg 1$. This region is beyond the limit of validity for the concept of parallel viscosity; however, the coefficients in (B5) do have a well-defined limit when $\tilde{\omega} \tau_{ii} \rightarrow \infty$,

$$a_1 + \frac{2p}{B^2} = \frac{\rho}{B^2} \left(\frac{9}{4} v_s^2 - \frac{1}{4} V_p^2 \right),$$

$$a_2 = \frac{1}{2}, \quad a_4 = 0. \quad (\text{B6})$$

Now, the sound wave has been *completely* removed, but the imaginary parts are still lacking, and the dependence on the parallel phase velocity is quite different from that in GCP. The plasma has effectively been made more rigid, and only displacements with isotropic compression, giving $\delta p / \rho = (3/2) \delta B / B$, are allowed. What used to be the slow wave continuum, $K = 1 + a_1 + 2p/B^2 = 0$, has moved to larger phase velocity, $V_p^2 = 4v_A^2 + 9v_s^2$. This is similar to the case of incompressible MHD, $\gamma = \infty$, where the slow wave continuum coincides with the Alfvén continuum at $V_p^2 = v_A^2$.

APPENDIX C: SUYDAM CRITERIA IN GCP, CGL, AND MHD FOR SHEARLESS FLOWS

For shearless flows ($M = 0$), the kinetic integrals in the Suydam criterion $D_0 < 1/4$ are all evaluated at zero phase velocity, and the coefficients a_1 – a_4 can be taken from Table I. The Suydam index (29) is split in a natural way as $D_0 = D_{\text{fluid}} + D_{\text{kin}}$, where D_{fluid} is independent of the kinetic integrals and has the same value in all the models, while

$$D_{\text{kin}} = \left(\frac{q}{r q' B_z} \right)^2 \frac{1}{\sigma B_\theta^2} \left(\frac{H^2}{K} - B^2 a_4 \rho v_\theta^4 \right) \quad (\text{C1})$$

is model dependent. As already noted, MHD with $\gamma = 1$ gives the same Suydam criterion as GCP for a distribution in local thermodynamic equilibrium. This was shown in the static case $M = V = 0$ by Pao.¹⁴ To compare the GCP and CGL conditions for anisotropic pressures, we note the inequalities

$$H^{\text{CGL}} = \sigma B_\theta^2 - \rho v_\theta^2 \left(1 - \frac{p_\perp}{3p_\parallel} \right) < H^{\text{GCP}}$$

$$= \sigma B_\theta^2 - \rho v_\theta^2 \left(1 - \frac{p_\perp}{p_\parallel} \right),$$

$$\begin{aligned}
K^{\text{CGL}} &= 1 + \frac{2p_{\perp}}{B^2} - \frac{p_{\perp}^2}{3p_{\parallel} B^2} > K^{\text{GCP}} \\
&= 1 + \frac{2p_{\perp}}{B^2} - \frac{2p_{\perp}^2}{p_{\parallel} B^2}, \\
-a_4^{\text{CGL}} &= \rho/3p_{\parallel} < -a_4^{\text{GCP}} = \rho/p_{\parallel}.
\end{aligned}$$

Consequently (assuming $H^{\text{GCP}} > 0$ and $K^{\text{GCP}} > 0$), each term in D_{kin} is smaller for CGL than for GCP, i.e., $D_0^{\text{CGL}} < D_0^{\text{GCP}}$ term by term, and the CGL equations are overly optimistic for isotropic, as well as for anisotropic equilibria. This also holds in the static case, $v_{\theta} = 0$, in agreement with the more general comparison between MHD and CGL given by Bernstein *et al.*²³ It appears that the double adiabatic approximation makes the plasma too rigid in the longitudinal direction by not allowing the temperatures to equilibrate along the field lines. Isothermal conditions are simply accounted for in MHD by setting $\gamma = 1$.

The MHD system with an infinite parallel viscosity is even more rigid than CGL. Evaluating the CGL coefficients for an isotropic equilibrium, we have the following comparison:

$$\begin{aligned}
H^{\text{MHD}}(\mu_{\parallel} = \infty) &= \sigma B_{\theta}^2 - \frac{3}{2} \rho v_{\theta}^2 < H^{\text{CGL}} = \sigma B_{\theta}^2 - \frac{3}{2} \rho v_{\theta}^2, \\
K^{\text{MHD}}(\mu_{\parallel} = \infty) &= 1 + \frac{3}{2} (\gamma p/B^2) > K^{\text{CGL}} = 1 + 5p/3B^2, \\
-a_4^{\text{MHD}}(\mu_{\parallel} = \infty) &= 0 < -a_4^{\text{CGL}} = \rho/3p.
\end{aligned}$$

It is clear that $D_0^{\text{MHD}}(\mu_{\parallel} = \infty) < D_0^{\text{CGL}} < D_0^{\text{MHD}}(\mu_{\parallel} = 0) = D_0^{\text{GCP}}$ for plasmas in local thermodynamic equilibrium, i.e., MHD with infinite parallel viscosity makes the plasma even more rigid than the double-adiabatic model.

Finally, we give explicitly the GCP Suydam index for isotropic equilibria with $M = 0$,

$$\begin{aligned}
D_0^{\text{GCP}} &= \left(\frac{q}{q' B_z} \right)^2 \left\{ -\frac{2p'}{r} + \frac{2\rho v_{\theta}^2}{r^2} + \frac{B^2}{B_{\theta}^2} \right. \\
&\quad \times \left[\frac{\rho^2 v_{\theta}^4}{r^2 p} - r \frac{d}{dr} \left(\frac{\rho v_{\theta}^2}{r^2} \right) \right] \Bigg\}, \quad (\text{C2})
\end{aligned}$$

which shows destabilization by the azimuthal flow as well as by an outward decrease in $\rho(v_{\theta}/r)^2$.

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